



Food and Agriculture Organization
of the United Nations

Anticipatory Cash Transfers in Somalia: Evidence on Timing, Modality, and Impact

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Motivation

Handful of rigorous evidence on the effectiveness

Expected increase in AA usage with improved forecasts

- The impact evaluations of AA remain fragmented and insufficient. (Weingärtner et al., 2020)
- As forecasting systems improve ...rigorous evidence on the effectiveness of AA is urgent (Christian et al., 2026)
- The delivery modalities also matters. We know little on which form of AA works best



Context

- Fragile States Index (2024): most fragile country (decades of conflict and political instability)
- Notre Dame Global Adaptation Initiative's Vulnerability Index (2024) ranks Somalia 181 out of 187 countries, due to extreme exposure, sensitivity, and limited adaptive capacity
- IPC estimates 4 million people (21% of the population) in IPC Phase 3 or worse, and 1.7 million children under five projected to experience acute malnutrition in 2024, with 430,000 severely malnourished
- ENSO forecasts (2024) suggest a 66% chance of La Niña in late 2024, associated with dry conditions and heightened drought risk in Somalia

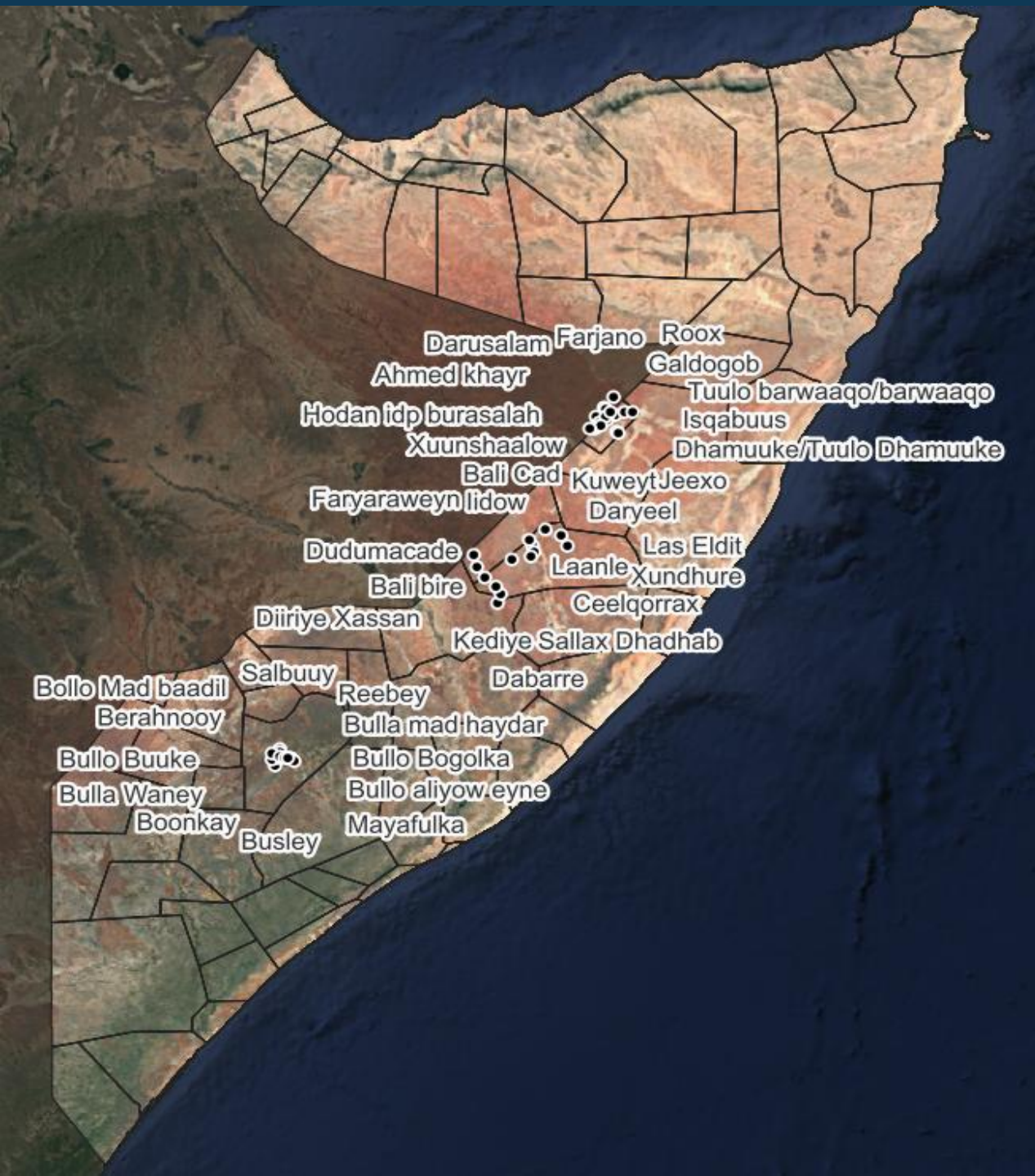


Research Questions

1. Is it more effective to deliver AA cash transfers as a lump-sum or through multiple instalments? (RCT-DiD) Lump-sum: USD 210 once | Multiple Instalments: USD 70/month × 3 months
2. What is the impact of receiving an anticipatory cash transfer on food security and livelihoods in Somalia? (IPW-DiD — supporting context)

Theory of Change





3 districts:

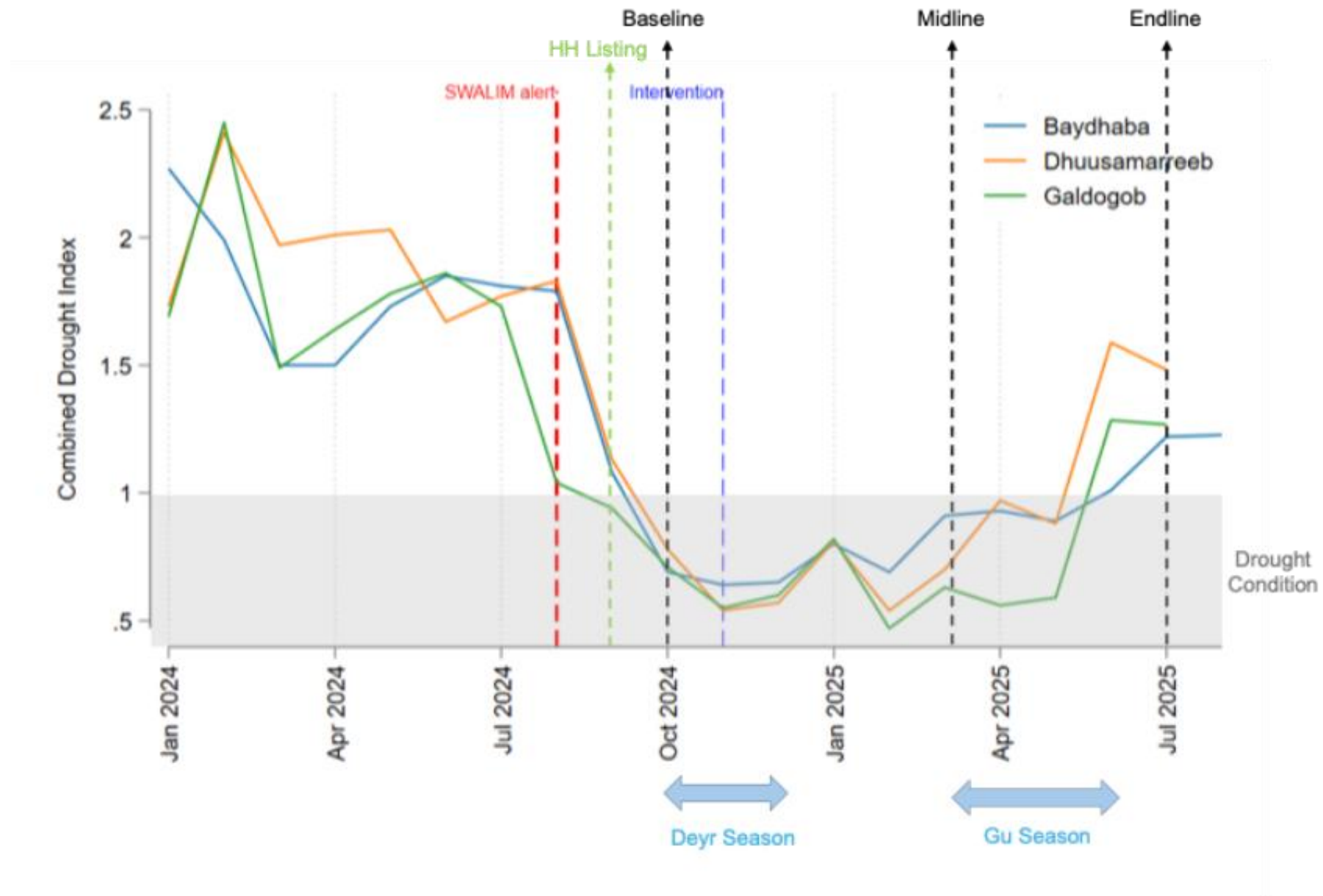
Baidoa

IPW-DiD --> Impact of EWBA

Dhusamareeb & Galdogob

RCT-DiD --> Lump-sum vs
multiple instalments

44 villages and 3,200
households over 3 waves



Research Designs

IPW-DiD

- 12 villages in Baidoa and approximately 1600 HHs
- Beneficiaries received a cash transfer of USD 240 in different modalities
- The IPW sample strongly balanced on baseline characteristics.

RCT-DiD

- 6 villages in Dhusamareeb and 3 villages in Galdogob and approximately 800 HHs
- Villages randomly selected into lump-sum and three instalments. HHs in lump-sum villages received USD 210 in November. HHs in instalment villages received 3 tranches of 70 USD in November, December and January
- Baseline characteristics balanced, good statistical power
- Wild cluster bootstrapped SE(10,000 iterations) and randomized inference (placebo permutation tests)

Empirical Strategy

- 1) Inverse Probability Weighted (IPW) Difference-in-Differences (DID) with pure controls (IPW-DiD)
- 2) Difference-in-Differences (DID) for Cluster-stratified Randomized Controlled Trial (RCT-DiD)

$$Y_{it} = \beta_0 + \beta_1 Treat_{it} + \beta_2 Post + \beta_3 Treat_{it} * POST + \eta X'_{it} + e_{it}$$

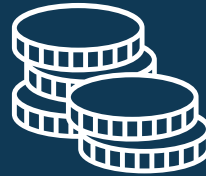
- Y_{it} is the outcome of interest for household i in survey round t ,
- $Treat (1)_{it}$ is a binary indicator equals 1 for treated households in the post-treatment periods (midline or endline),
- $Treat (2)_{it}$ is a binary indicator equals to 1 if households receive the cash transfer in one lump sum solution and 0 if the cash transfer is delivered in three instalments
- β_3 is the ATE of anticipatory cash transfers on the outcome Y
- X_{it} is a vector of time-varying household-level controls: wealth index, indicators for receiving wage income, other cash or in-kind assistance, and whether the survey round occurred during Ramadan, and
- e_{it} is an iid error term.
- IPW weights are $1/\hat{p}(X_i)$ for treated and $1/(1-\hat{p}(X_i))$, for control households. $\hat{p}(X_i)$ is the predicted propensity score of receiving treatment conditional on observed baseline characteristics (household malnourishment, disability, chronic illness, age and gender of the household head, engagement in agriculture, and total livestock holdings)

Main findings

33%
increase in FCS



66% more have
crop reserves; 7%
less likely face
crop losses



Savings increased
Debt repayment
increased



Reduction in
migrations and
displacements



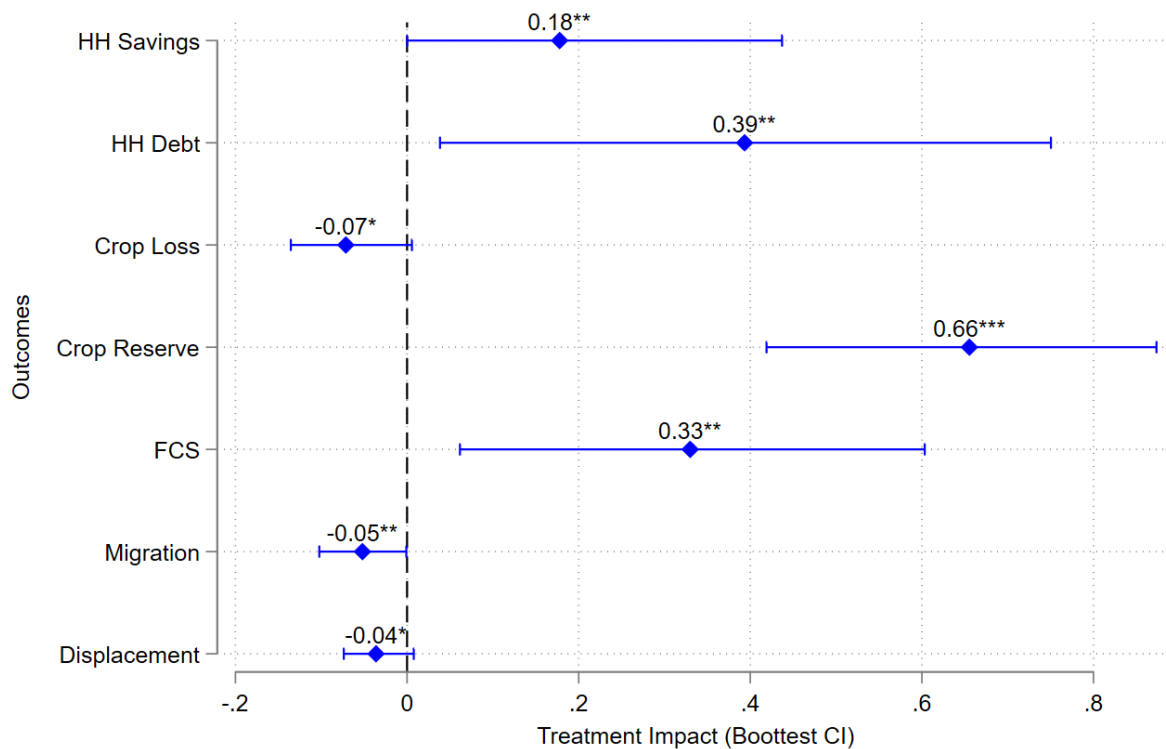
Livestock losses
reduced for lump-
sum households

IPW-DiD

RCT-DiD

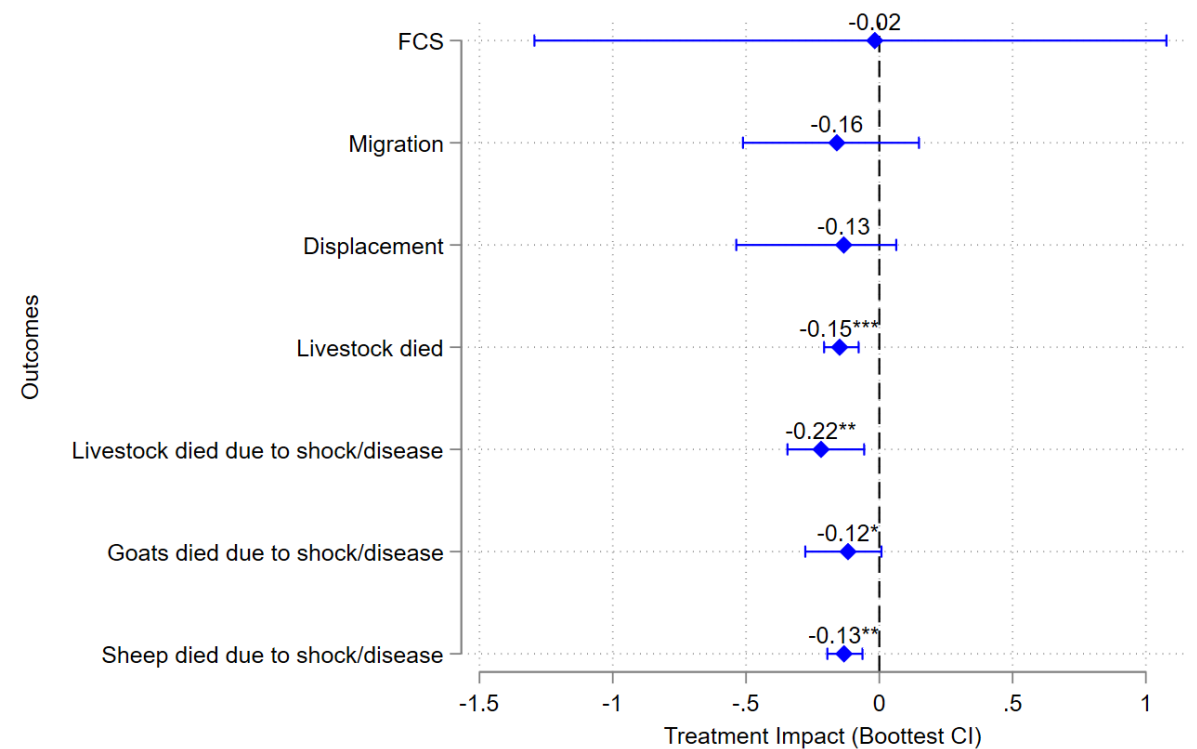
Results Midline

IPW-DiD: Treatment → EWBA cash



* p < .1, ** p < .05, *** p < .01, **** p < .001

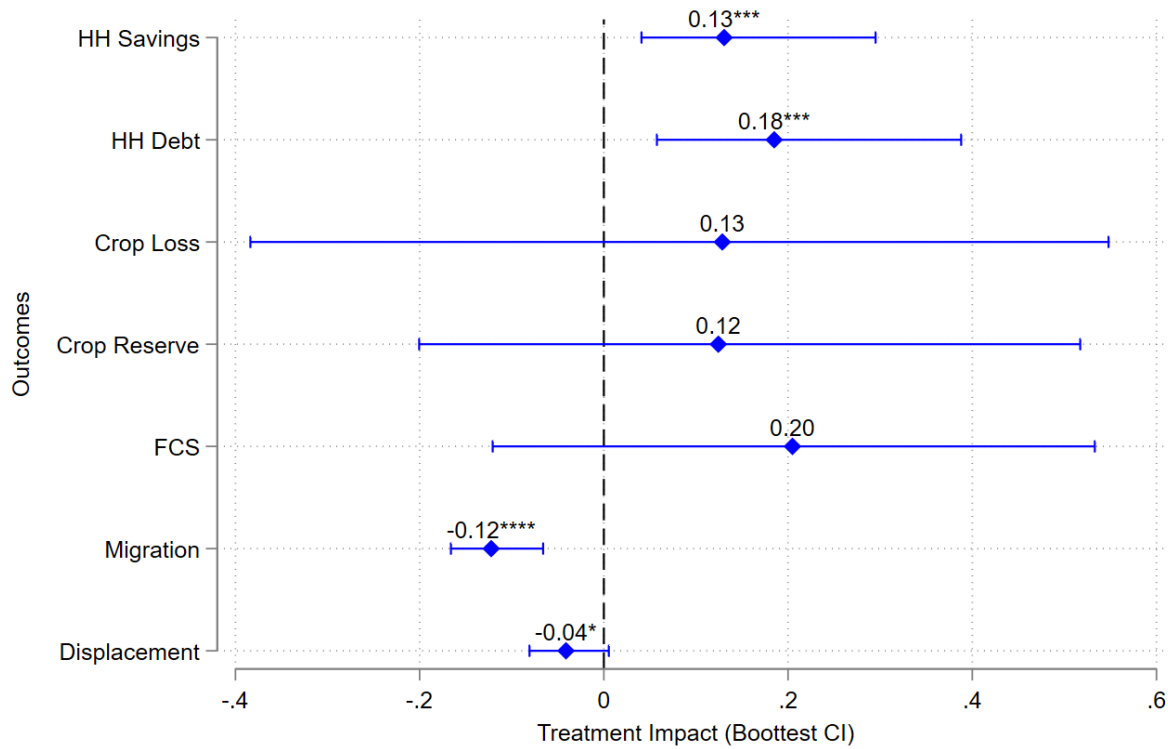
RCT-DiD: Treatment → Lump-sum



* p < .1, ** p < .05, *** p < .01, **** p < .001

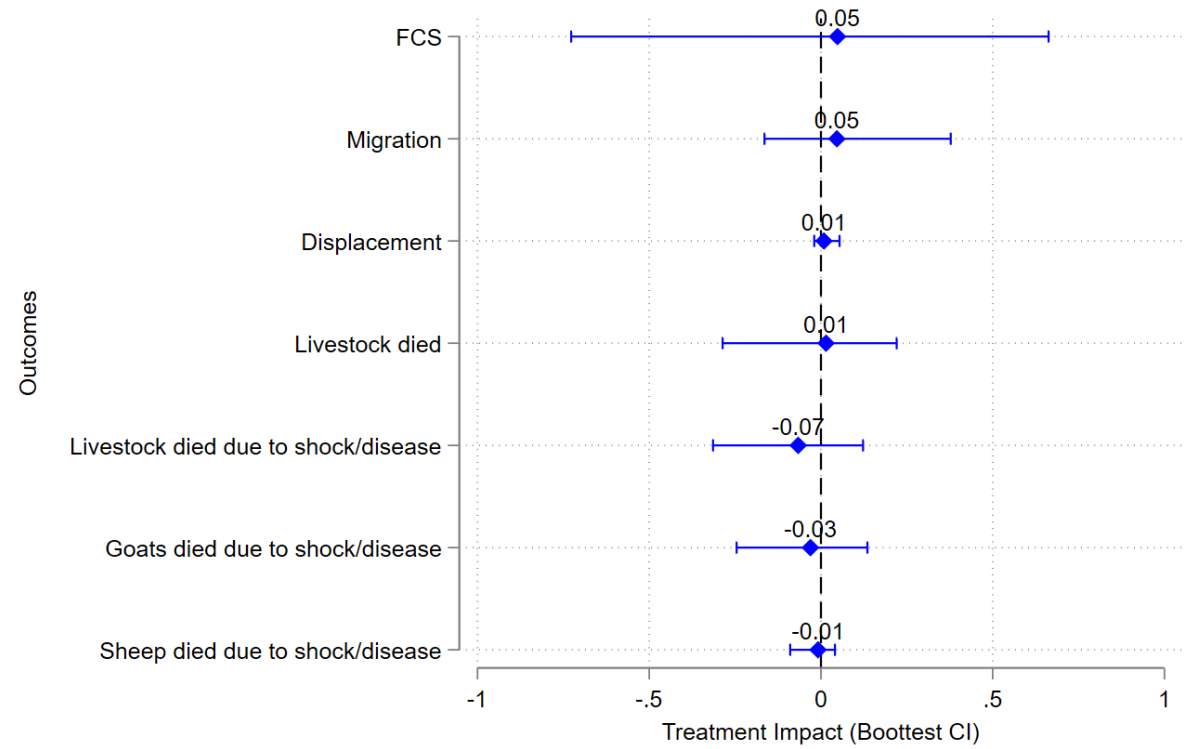
Results Endline

IPW-DiD: Treatment → EWBAA cash



* p < .1, ** p < .05, *** p < .01, **** p < .001

RCT-DiD: Treatment → Lump-sum



* p < .1, ** p < .05, *** p < .01, **** p < .001

Policy Implications & Conclusion

1

Early support delivers more:

Providing assistance before the crisis peak better protects food security, limits displacement, and preserves financial stability.

2

One payment beats multiple instalments:

Lump-sum cash transfers outperform staggered instalments in shielding productive assets in the face of shock, particularly livestock.

3

Short-term relief, long-term gains:

Anticipatory cash transfers help households absorb shocks and show early signs of contributing to lasting outcomes.

4

Gains don't always last:

Impact on several outcomes fades by endline, highlighting the importance of complementary interventions to sustain gains over time.



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Thank you

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Appendix

Power Calculations at Baseline

Outcome Variable	Mean	SD	ICC	MDE
HH Savings	0.055	0.228	0.051	0.089
HH Debt Paid	0.256	0.437	0.016	0.108
Any crop loss	0.920	0.272	0.002	0.042
Crop reserve	0.270	0.444	0.154	0.288
FCS	0.453	0.187	0.045	0.069
Anyone migrate	0.043	0.202	0.031	0.064
Total TLU	0.033	0.075	0.021	0.020
Died TLU	0.030	0.090	0.024	0.026

Variable	Mean	SD	ICC	MDE
FCS	0.628	0.182	0.227	0.166
TLU Died Animal	0.159	0.175	0.304	0.184
TLU Died shock and disease	0.161	0.208	0.285	0.212
Anyone Migrate	0.028	0.166	0.029	0.063

Important Literature

Bangladesh

Gros et al. (2019, 2023): + Evacuate homes, - Selling valuable household assets or taking new loans, - reduced meal-skipping behavior and - perceived stress agricultural losses and health issues

Pople et al. (2021): + Adult and child FCS, - in asset loss and costly borrowing, with an overall + household wellbeing

Mongolia

Gros et al. (2020): inconsistent effects on offspring survival rates for goats and sheep

Mogge et al. (2024): - livestock mortality and + herd-related investments and + livestock consumption

Nigeria

Balana et al. (2023): + positive effects of AA, although no significant changes were observed in FCS

Nepal

Dunsch et al. (2024): + effects on FCS, subjective wellbeing, and psychological distress, but no significant impact was found on agricultural outcomes, borrowing, or non-food expenditures

**Additional new literature in this sphere from Pople et al. 2026 and Baliki et al. 2026 from Niger and South Sudan, respectively, are not listed*

Robustness Checks: Modality RCT (RI Test)

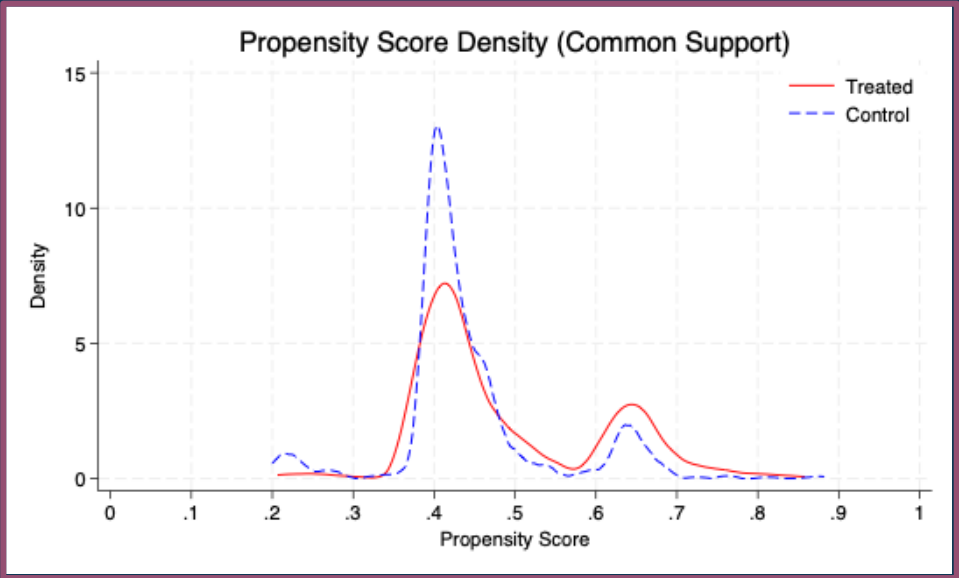
Variable	CI lower	CI upper	SE	p-value	Coefficient	Repetitions
Midline						
FCS	0.953	0.976	0.006	0.966	-0.017	1000
Died TLU	0.018	0.039	0.005	0.027**	-0.149	1000
Died due to shock/disease	0.021	0.044	0.005	0.031**	-0.219	1000
Any HH member migrate	0.317	0.376	0.015	0.346	-0.159	1000
Goats died due to shock/disease	0.101	0.143	0.010	0.121	-0.117	1000
Sheep died due to shock/disease	0.027	0.052	0.006	0.038**	-0.132	1000
Big animals died due to shock/disease	0.749	0.801	0.013	0.776	0.039	1000
Endline						
FCS	0.799	0.847	0.012	0.824	0.048	1000
Died TLU	0.819	0.865	0.012	0.843	0.015	1000
Died due to shock/disease	0.412	0.474	0.016	0.443	-0.066	1000
Any HH member migrate	0.761	0.813	0.013	0.788	0.046	1000
Goats died due to shock/disease	0.626	0.685	0.015	0.656	-0.031	1000
Sheep died due to shock/disease	0.736	0.790	0.013	0.764	-0.009	1000
Big animals died due to shock/disease	0.371	0.433	0.016	0.402	-0.127	1000

RCT Regressions

	(1)	(2)	(3)	(4)		
	FCS	TLU of all died animals	TLU of died animals due to shock and disease	Anyone migrated since last survey round	TLU of goats died due to shock and disease	TLU of sheep died due to shock and disease
Midline						
Lump-sum* Midline	-0.017	-0.149***	-0.260**	-0.159	-0.117*	-0.132**
	(0.458)	(0.029)	(0.059)	(0.125)	(0.057)	(0.027)
N	1520	1168	1168	1520	1150	708
R ²	0.174	0.128	0.135	0.152	0.072	0.129
Boottest p-value	ns	***	**	ns	*	**
RI Test p -value	0.966	0.027**	0.031	0.346	0.121	0.038**
Time FE	✓	✓	✓	✓	✓	✓
HH FE	✓	✓	✓	✓	✓	✓
Endline						
Lump-sum * Endline	0.048	0.015	-0.066	0.046	-0.031	-0.009
	(0.281)	(0.094)	(0.084)	(0.102)	(0.072)	(0.024)
N	1547	1201	1201	1547	1183	724
R ²	0.033	0.153	0.107	0.069	0.042	0.051
Boottest p-value	Ns	Ns	ns	ns	ns	ns
RI Test p- value	0.824	0.843	0.443	0.788	0.656	0.764
Time FE	✓	✓	✓	✓	✓	✓
HH FE	✓	✓	✓	✓	✓	✓

Robustness Checks: Quasi Experimental (IPW)

	Pre-Balance			Post-Balance		
	Treated	Untreated	p-value	Treated	Untreated	p-value
Any HH member malnourished	0.120	0.130	0.486	0.130	0.130	0.790
HH member disabled	0.100	0.100	0.552	0.100	0.100	0.919
HH member chronic illness	0.030	0.070	0.001***	0.050	0.050	0.808
HHH Age	37.810	37.080	0.256	37.520	37.460	0.938
HHH Female	0.530	0.530	0.996	0.530	0.520	0.870
HH does Agriculture	0.750	0.890	0.000***	0.820	0.830	0.985
Tropical Livestock Unit	0.890	0.660	0.001***	0.770	0.820	0.644



Quasi-experimental Regressions

	(1)	(2)	(3)	(4)	(5)	(6)
	HH saved since last wave	HH paid any debt since last wave	HH suffered any crop loss (in last ag season)	HH has crop reserves (at the time of interview)	Food Consumption Score	Anyone migrated since last survey round
Midline						
Treatment * Midline	0.178**	0.393**	-0.065**	0.655***	12.338*	-0.052**
	(0.098)	(0.148)	(0.024)	(0.098)	(6.193)	(0.020)
N	3214	3214	3214	3214	3214	3214
R ²	0.087	0.161	0.098	0.229	0.267	0.070
Boottest p-value	**	**	**	***	*	**
Boottest CI	[.0001944, .437]	[.03835, .7504]	[-.1205, -.008494]	[.4189, .8734]	[-2.175, 27.18]	[-.1023, -.0008607]
Time FE	✓	✓	✓	✓	✓	✓
HH FE	✓	✓	✓	✓	✓	✓
Endline						
Treatment * post=1	0.131**	0.185**	0.059	0.124	6.405	-0.122
	(0.052)	(0.064)	(0.134)	(0.140)	(5.802)	(0.019)
N	3235	3235	3235	3235	3235	3235
R ²	0.114	0.150	0.083	0.113	0.149	0.096
Boottest p-value	***	***	ns	ns	ns	***
Boottest CI	[.04086, .295]	[.05762, .3879]	[-.3361, .3831]	[-.2005, .5172]	[-7.74, 19.51]	[-.166, -.06592]
Time FE	✓	✓	✓	✓	✓	✓
HH FE	✓	✓	✓	✓	✓	✓